

## Chapter 2 : Introduction

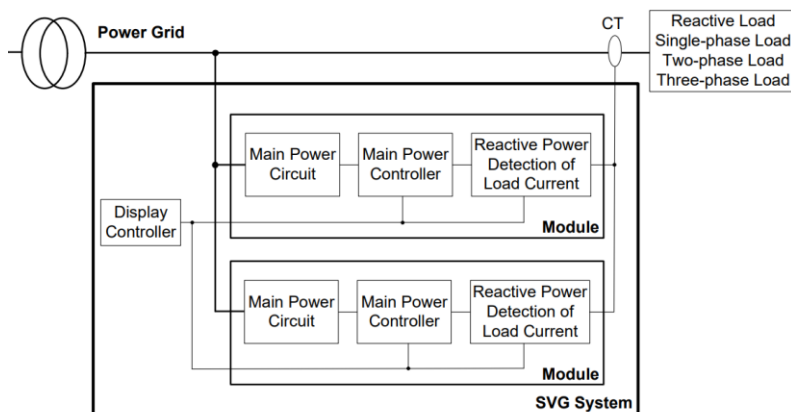
### 2.1 Product Introduction

The Delta PQC Series Mixed Fixed type SVG, a reactive power compensator for the three-phase power grid, is applied to the reactive power control of the power grid. Features of the device are shown as below:

- Compensates both inductive (lagging) and capacitive (leading) reactive power.
- Compensates 2nd-25th harmonic currents(selectable) with 20% of rated current output capacity, 100kvar SVG can output 30A H2-H25 harmonic compensation current.
- Correct three-phase load imbalance.
- Rapid dynamic responses, stable parameters and good harmonic compensation results.
- High efficiency and low loss. The advanced ECO mode allows you to save more energy.
- The modular design allows you to freely select different modules for the harmonic power compensation for different capacities.
- The system adopts an advanced 3-level structure and consists of digital signal processors (DSP), Field Programmable Gate Array (FPGA) and high power electronic devices, which has excellent performance and superior reliability.
- Supports remote on/off and fault monitoring via output relays and supports Modbus RTU/ TCP for remote SCADA monitoring.

Please see **Figure 2-1** for the SVG system block diagram.

The SVG system consists of multiple SVG modules and one system cabinet. Load current is detected by external CT and load reactive power is calculated by SVG controller. Based on the load current feedback, the controller manages power circuit to output inverse reactive power to cancel out load reactive power. The system cabinet is equipped with a HMI Display that shows the information of individual modules and control the operation of individual modules. All power modules are paralleled connection and they are independently working.



(Figure 2-1: SVG System Block Diagram)